

System Architecture, Deterministic Engines & Technical Specification

A Native Amazon Bedrock Platform for Wealth Advisory & Independent RIA Operations

Target Cloud: Amazon Web Services (AWS)	Active Region: us-east-1 (N. Virginia)
Bedrock Model: Amazon Nova Pro (us.amazon.nova-pro-v1:0)	SDK Protocol: Native boto3 Converse API (Zero LangChain)
System Date: October 03, 2026	Identity & Security: IAM Role / STS Caller Identity

Executive Summary: Briefly is an enterprise AI copilot designed for wealth advisors. Unlike generic LLM wrappers that fabricate calculations, Briefly enforces a strict architectural boundary: **deterministic calculations** (RMDs, portfolio drift, tax-loss harvesting, wash-sale detection) are executed strictly in verifiable local Python code, while **Amazon Bedrock** operates as a supervisor agent via its native Converse API. Bedrock Guardrails filter out non-compliant advice, and an advisor approval gate ensures no email, trade, or CRM alteration occurs without explicit human authorization.

1. System Architecture & Component Isolation

Briefly relies exclusively on native AWS SDK primitives. The architecture comprises five isolated layers:

Layer	AWS Service / Component	Function & Operational Security Boundary
1. Presentation	Streamlit Dashboard	Advisor workspace with client switcher, metrics snapshot, task tracks, meeting brief export, and high-contrast LPL Assistant chat.
2. Agentic Supervisor	Amazon Bedrock Converse API (us.amazon.nova-pro-v1:0)	Model-driven autonomous reasoning loop. Decides when to invoke financial tools, inspects arguments, and synthesizes grounded natural language briefs.
3. Safety & Policy	Amazon Bedrock Guardrails & Bedrock Knowledge Bases	Inspects prompts and outputs against fiduciary constraints (PII redaction, investment advice disclaimers). Retrieves firm compliance documents from S3.
4. Calculation Core	Deterministic Planning Engine (planning.py)	IRS Pub. 590-B Uniform Lifetime RMD tables, lot-level portfolio rebalancing math, wash-sale detection, and SEC Reg BI meeting documentation screens.
5. Governance	Durable SQLite Audit Trail (audit.py)	Immutable session log capturing proposed advisor actions, reviewer identity, UTC timestamp, and approval/rejection decisions.

2. Deterministic Financial Planning Specification (planning.py)

A foundational principle of Briefly is that **large language models must never compute client financial balances or tax liabilities**. The module `planning.py` implements pure mathematical business logic across four critical advisor workflows:

2.1 Required Minimum Distribution (RMD) Planner — SECURE 2.0 Compliance

Under the SECURE 2.0 Act, required beginning ages have transitioned based on birth cohort. Briefly automatically categorizes pre-tax retirement accounts (Traditional IRA, SEP, SIMPLE, Rollover, 401k) and divides the prior year-end balance by the IRS Uniform Lifetime Table (Publication 590-B) factor:

Birth Year Cohort	SECURE 2.0 Start Age	Distribution Deadline	Statutory Rule & Penalty Avoidance
Born 1950 or earlier	Age 72	December 31 of current year	Original SECURE Act 1.0 schedule.
Born 1951 – 1959	Age 73	April 1 following year (1st yr), Dec 31 thereafter	SECURE 2.0 primary transition band.
Born 1960 or later	Age 75	April 1 following year (1st yr), Dec 31 thereafter	SECURE 2.0 extended horizon for younger retirees.

2.2 Lot-Level Allocation Drift & Concentration Limits

Holdings are aggregated across taxable, Traditional IRA, and Roth IRA sub-accounts. Asset classes (Equity, Fixed Income, Cash) are measured against firm model profiles. When any asset class diverges by more than **±5.0 percentage points**, an alert is triggered. Single-stock positions exceeding **10.0%** are isolated for concentration risk review.

Risk Profile	Target Equity	Target Fixed Income	Target Cash	Rebalance Action Trigger
Aggressive	80.0%	15.0%	5.0%	Equity allocation <75% or >85%
Moderate	60.0%	35.0%	5.0%	Equity allocation <55% or >65%
Conservative	30.0%	60.0%	10.0%	Equity allocation <25% or >35%

2.3 Tax-Loss Harvesting & Wash-Sale Screening (IRC § 1091)

Only taxable lots are evaluated for tax-loss opportunities. The screener applies a minimum \$500 loss threshold, classifies short-term vs. long-term holding periods, and scans cross-account purchases within a **30-day window** to prevent IRS Section 1091 wash-sale disallowance. Non-substantially-identical replacement proxies (e.g. S&P; 500 to Total Stock Market) are recommended.

2.4 SEC Regulation Best Interest (Reg BI) Meeting Documentation Screen

Prior to committing meeting notes to a CRM or compliance file, Briefly runs a 10-point audit checklist against raw transcripts:

Reg BI Audit Element	Evaluation Focus	Standard Required Evidence
1. Client Goals	Stated retirement or wealth milestones	Target retirement date, capital growth objective
2. Risk Profile	Risk capacity & tolerance to drawdown	Confirmation of volatility willingness
3. Time Horizon	Liquidity horizon and life milestones	Estimated withdrawal timeframe in years
4. Cash Flow & Liquidity	Immediate or near-term capital needs	Emergency reserve amounts or planned expenditures
5. Tax Considerations	Tax status and distribution strategy	Marginal tax bracket, Roth vs. Traditional timing
6. Fee Disclosures	Advisory, custodial, and expense ratios	Explicit presentation of product costs
7. Alternatives Analyzed	Comparison of competing investment options	Documentation of why alternative was rejected
8. Conflict Disclosures	Affiliated compensation or incentives	Form ADV Part 2A / Form CRS delivery
9. Client Consent	Explicit advisor agreement	Verbal or written client concurrence
10. Actionable Next Steps	Assigned owners, tasks, and follow-ups	Clear due dates for execution

3. Amazon Bedrock Orchestration & Converse Loop

Briefly executes on Amazon Bedrock's native Converse API with Amazon Nova Pro (us.amazon.nova-pro-v1:0). This architecture delivers predictable token utilization, exact JSON schema validation, and zero framework overhead.

```
// Bedrock Converse Native Tool Loop (agent.py) request = { 'modelId': 'us.amazon.nova-pro-v1:0', 'system': [{ 'text': SYSTEM_PROMPT}], 'messages': messages, 'toolConfig': self.tool_config, 'inferenceConfig': { 'maxTokens': 1200, 'temperature': 0.2} } response = bedrock_runtime.converse(**request)
```

3.1 Implemented Action Tools

Tool Identifier	Input Schema	Operational Description
get_portfolio_drift	client_json: object	Computes allocation divergence vs. target model. Returns percentage point delta.
query_crm_notes	client_id: str, query: str	Natural language keyword retrieval across prior advisor CRM interaction logs.
draft_compliance_log	action: str, details: str	Intercepts external action requests (email, trades). Formats payload for approval gate.
query_book_metrics	query_type: enum	Aggregates book-level metrics (AUM, client count, urgent tasks, pending deadlines).
search_advisor_library	query: str	Vector RAG search over Bedrock Knowledge Base connected to S3 compliance manuals.

4. Fiduciary Governance, Audit Trail & Security Boundary

In wealth management, autonomous agent execution presents unacceptable regulatory risk. Briefly enforces an air-gap between model synthesis and production actions:

4.1 Human-in-the-Loop Fiduciary Gate

Whenever the supervisor model determines that an outbound email, CRM update, or trade should be performed, it emits a BRIEFLY_PROPOSAL envelope. Execution is immediately halted. The proposal surfaces in the Streamlit UI with **Approve & Log** and **Reject** buttons. No external API is triggered without human approval.

4.2 Durable SQLite Audit Trail (audit.py)

All advisor approvals and rejections are immutably logged to an SQLite database with UTC timestamps, reviewer credentials, proposed action text, and client ID. The log can be exported directly to CSV for compliance review:

Field	Type	Description & Compliance Value
id	INTEGER PRIMARY KEY	Autoincrementing record sequence number.
reviewed_at	TEXT (ISO 8601)	UTC timestamp of the human advisor's final decision.
client_id	TEXT	Stable synthetic client identifier (e.g. C-1001).
action	TEXT	Categorical action requested (e.g. send_email, update_crm).
details	TEXT	Verbatim proposed action payload drafted by Bedrock.
decision	TEXT	Explicit advisor ruling: 'approved' or 'rejected'.
reviewer	TEXT	Authenticated user identity (IAM role / advisor ID).